



Bahria University, Islamabad
Department of Software Engineering

Machine Learning Lab
(Fall-2026)

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Enrollment: 01-131232-105

Lab Journal: 02

Date: 08-09-2026

Task No:	Task Wise Marks		Documentation Marks		Total Marks (20)
	Assigned	Obtained	Assigned	Obtained	
1	3		5		
2	3				
3	3				
4	3				
5	3				

Comments:

Signature

Demonstrate how to perform exploratory data analysis on give dataset

```
[1] import pandas as pd
import matplotlib.pyplot as plt

# Superstore dataset
df = pd.read_csv('Superstore.csv', encoding='latin1')

# Print size (rows and columns)
print("Dataset Shape (Rows, Columns):", df.shape)

# Print basic info about variable types
print("\n DATA INFO ")
df.info()

# 3. Calculate summary statistics.
print("\n SUMMARY STATISTICS ")
print(df.describe())
```

Dataset Shape (Rows, Columns): (9994, 21)

--- DATA INFO ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
Column Non-Null Count Dtype

0 Row ID 9994 non-null int64
1 Order ID 9994 non-null object
2 Order Date 9994 non-null object
3 Ship Date 9994 non-null object
4 Ship Mode 9994 non-null object
5 Customer ID 9994 non-null object
6 Customer Name 9994 non-null object
7 Segment 9994 non-null object
8 Country 9994 non-null object
9 City 9994 non-null object
10 State 9994 non-null object
11 Postal Code 9994 non-null int64
12 Region 9994 non-null object
13 Product ID 9994 non-null object
14 Category 9994 non-null object
15 Sub-Category 9994 non-null object
16 Product Name 9994 non-null object
17 Sales 9994 non-null float64
18 Quantity 9994 non-null int64
19 Discount 9994 non-null float64
20 Profit 9994 non-null float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB

--- SUMMARY STATISTICS ---

	Row ID	Postal Code	Sales	Quantity	Discount
count	9994.000000	9994.000000	9994.000000	9994.000000	9994.000000
mean	4997.500000	55190.379428	229.858001	3.789574	0.156203
std	2885.163629	32063.693350	623.245101	2.225110	0.206452
min	1.000000	1040.000000	0.444000	1.000000	0.000000
25%	2499.250000	23223.000000	17.280000	2.000000	0.000000
50%	4997.500000	56430.500000	54.490000	3.000000	0.200000
75%	7499.750000	90008.000000	209.940000	5.000000	0.200000
max	9994.000000	99301.000000	22638.480000	14.000000	0.800000

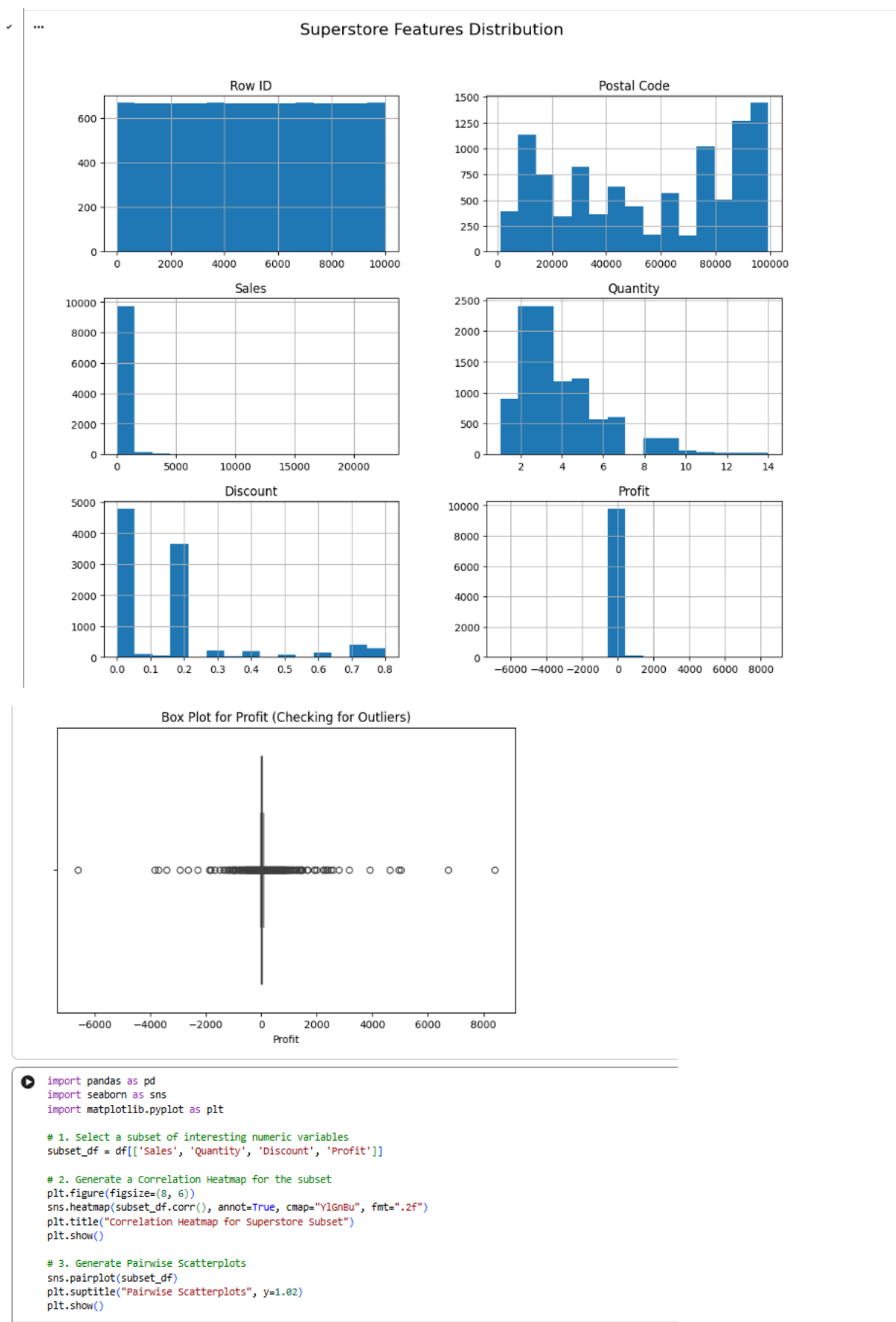
Profit

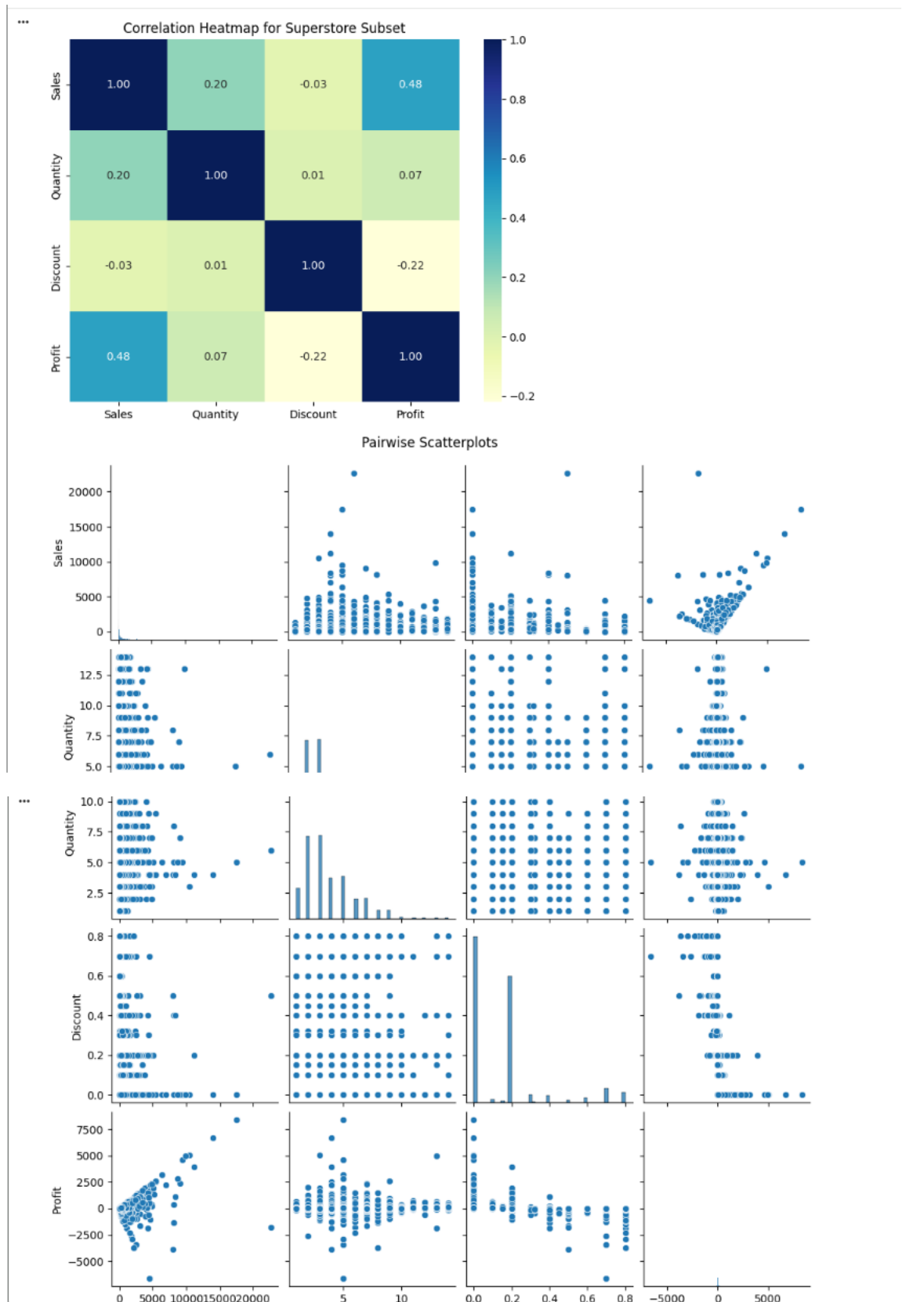
count	9994.000000
mean	28.656896
std	234.260108
min	-6599.978000
25%	1.728750
50%	8.666500
75%	29.364000
max	8399.976000

```
import seaborn as sns
import matplotlib.pyplot as plt

# Create histograms for numeric variables to check distribution
df.hist(figsize=(12, 10), bins=15)
plt.suptitle("Superstore Features Distribution", fontsize=16)
plt.show()

# Create a box plot to check for outliers in Sales or Profit
plt.figure(figsize=(8, 5))
sns.boxplot(x=df['Profit'])
plt.title("Box Plot for Profit (Checking for Outliers)")
plt.show()
```





Thank you
